

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### Product Identifier

**Product Description:** Glycine  
**Cat No. :** **S60281**  
**Synonyms** Gly; Amino acetic acid  
**CAS No** 56-40-6  
**Molecular Formula** C2 H5 N O2

### Relevant identified uses of the substance or mixture and uses advised against

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

### Details of the supplier of the safety data sheet

|                                          |                                          |
|------------------------------------------|------------------------------------------|
| <b>Importer</b>                          | <b>Supplier</b>                          |
| Fisher Scientific Korea                  | Thermo Fisher Scientific Chemicals, Inc. |
| D5,D6, Incheon Airport Logistics Complex | 30 Bond Street                           |
| 150, Gonghangdong-Ro 296 Beon-Gil        | Ward Hill, MA 01835-8099                 |
| Jung-Gu, Incheon                         |                                          |
| Tel: +82-1661-9555                       |                                          |
| Fax: +82-2-2023-0603                     |                                          |

**E-mail address** Chem.KR@thermofisher.com

### Emergency Telephone Number

**Emergency telephone: Medical:** +(82) 070-7686-0086 or + 1-703-741-5970  
**CHEMTREC:** 080 822 1374 (Local), CHEMTREC: 1-800-424-9300 or + 1-703-527-3887  
**Korea:** 00-308-13-2549 (24 hours a day, 7 days a week)

## SECTION 2: HAZARDS IDENTIFICATION

### Classification of the substance or mixture

#### Physical hazards

Based on available data, the classification criteria are not met

#### Health hazards

Based on available data, the classification criteria are not met

#### Environmental hazards

Based on available data, the classification criteria are not met

### Label Elements

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## Other Hazards

This product does not contain any known or suspected endocrine disruptors

## NFPA

Health  
0

Flammability  
1

Instability  
1

Physical hazards  
N/A

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component | Common Name            | CAS No  | Index No | Weight % |
|-----------|------------------------|---------|----------|----------|
| Glycine   | Gly; Amino acetic acid | 56-40-6 | KE-01153 | 99 - 100 |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

|                                           |                                                                                                                         |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.         |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur. |
| <b>Ingestion</b>                          | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                   |
| <b>Inhalation</b>                         | Remove to fresh air. Get medical attention immediately if symptoms occur.                                               |
| <b>Self-Protection of the First Aider</b> | No special precautions required.                                                                                        |

### Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

### Indication of any immediate medical attention and special treatment needed

**Notes to Physician** Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

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## Hazardous Combustion Products

Nitrogen oxides (NO<sub>x</sub>), Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

## Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

### Environmental precautions

Should not be released into the environment.

### Methods and Material for Containment and Cleaning Up

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation. Avoid dust formation.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a cool, well-ventilated place. Keep under nitrogen.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component | CAS No  | Korea      | ACGIH TLV  | OSHA PEL   |
|-----------|---------|------------|------------|------------|
| Glycine   | 56-40-6 | Not listed | Not listed | Not listed |

| Component | CAS No  | European Union | The United Kingdom | Germany    |
|-----------|---------|----------------|--------------------|------------|
| Glycine   | 56-40-6 | Not listed     | Not listed         | Not listed |

### ACGIH - Biological Exposure Indices

| Component | CAS No  | ACGIH - Biological Exposure Indices |
|-----------|---------|-------------------------------------|
| Glycine   | 56-40-6 | Not listed                          |

### Exposure Controls

#### Engineering Measures

None under normal use conditions.

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## Personal protective equipment

### **Eye Protection**

Wear safety glasses with side shields (or goggles)

### **Hand Protection**

Protective gloves

### **Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

### **Personal protective equipment**

Use only those certified by the Korea Occupational Safety and Health Administration.

### **Respiratory Protection**

No protective equipment is needed under normal use conditions

### **Recommended Filter type:**

Particle filter

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

**Appearance (Physical State, Color, etc.)** White Powder Solid

### **Odor**

Odorless

### **Odor Threshold**

No data available

### **pH**

5.9-6.4 @ 20°C

5% aq.solution

### **Melting Point/Range**

233 °C / 451.4 °F

### **Softening Point**

No data available

### **Boiling Point/Range**

No information available

### **Flash Point**

No information available

**Method -** No information available

### **Evaporation Rate**

Not applicable

Solid

### **Flammability (solid,gas)**

No information available

### **Explosion Limits**

No data available

### **Vapor Pressure**

No data available

### **Vapor Density**

Not applicable

Solid

### **Specific Gravity / Density**

1.595

### **Bulk Density**

No data available

### **Water Solubility**

Soluble

### **Solubility in other solvents**

No information available

### **Partition Coefficient (n-octanol/water)**

| Component | CAS No | log Pow |
|-----------|--------|---------|
|-----------|--------|---------|

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|         |         |       |
|---------|---------|-------|
| Glycine | 56-40-6 | -3.21 |
|---------|---------|-------|

|                           |                          |       |
|---------------------------|--------------------------|-------|
| Autoignition Temperature  | Not applicable           |       |
| Decomposition Temperature | No data available        |       |
| Viscosity                 | Not applicable           | Solid |
| Explosive Properties      | No information available |       |
| Oxidizing Properties      | No information available |       |
| Molecular Formula         | C2 H5 N O2               |       |
| Molecular Weight          | 75.07                    |       |

## SECTION 10: STABILITY AND REACTIVITY

|                                           |                                                                                           |
|-------------------------------------------|-------------------------------------------------------------------------------------------|
| <u>Reactivity</u>                         | None known, based on information available                                                |
| <u>Chemical Stability</u>                 | Moisture sensitive.                                                                       |
| <u>Possibility of Hazardous Reactions</u> |                                                                                           |
| Hazardous Polymerization                  | No information available.                                                                 |
| Hazardous Reactions                       | No information available.                                                                 |
| <u>Conditions to Avoid</u>                | Excess heat. Incompatible products. Exposure to moist air or water. Avoid dust formation. |
| <u>Incompatible Materials</u>             | Bases. Oxidizing agent.                                                                   |
| <u>Hazardous Decomposition Products</u>   | Nitrogen oxides (NOx). Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ).           |

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### Information on expected route of exposure

|            |                                                                                 |
|------------|---------------------------------------------------------------------------------|
| Inhalation | May cause irritation of respiratory tract.                                      |
| Ingestion  | Ingestion may cause gastrointestinal irritation, nausea, vomiting and diarrhea. |
| Eyes       | May cause irritation.                                                           |
| Skin       | May cause irritation.                                                           |

#### Information on Health Hazards

##### (a) acute toxicity;

|            |                                                                  |
|------------|------------------------------------------------------------------|
| Oral       | Based on available data, the classification criteria are not met |
| Dermal     | No data available                                                |
| Inhalation | No data available                                                |

|           |        |           |             |                 |
|-----------|--------|-----------|-------------|-----------------|
| Component | CAS No | LD50 Oral | LD50 Dermal | LC50 Inhalation |
|-----------|--------|-----------|-------------|-----------------|

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|---------|---------|---------------------------|-------------------|-------------------|
| Glycine | 56-40-6 | LD50 = 7930 mg/kg ( Rat ) | No data available | No data available |
|---------|---------|---------------------------|-------------------|-------------------|

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;  
Respiratory No data available  
Skin No data available

| Component | CAS No  | Test method       | Test species      | Study result      |
|-----------|---------|-------------------|-------------------|-------------------|
| Glycine   | 56-40-6 | No data available | No data available | No data available |

(e) germ cell mutagenicity; No data available

| Component | CAS No  | Test method       | Test species      | Study result      |
|-----------|---------|-------------------|-------------------|-------------------|
| Glycine   | 56-40-6 | No data available | No data available | No data available |

(f) carcinogenicity; No data available

| Component | CAS No  | Test method       | Test species / Duration | Study result      |
|-----------|---------|-------------------|-------------------------|-------------------|
| Glycine   | 56-40-6 | No data available | No data available       | No data available |

There are no known carcinogenic chemicals in this product

| Component | CAS No  | IARC       | NTP        | ACGIH      | OSHA       | UK         |
|-----------|---------|------------|------------|------------|------------|------------|
| Glycine   | 56-40-6 | Not listed | Not listed | Not listed | Not listed | Not listed |

(g) reproductive toxicity; No data available

| Component | CAS No  | Test method       | Test species / Duration | Study result      |
|-----------|---------|-------------------|-------------------------|-------------------|
| Glycine   | 56-40-6 | No data available | No data available       | No data available |

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs No information available.

(j) aspiration hazard; Not applicable  
Solid

Other Adverse Effects  
No information available.

| Component | CAS No  | EU - Endocrine Disrupters Candidate List | EU - Endocrine Disruptors - Evaluated Substances | Japan - Endocrine Disruptor Information |
|-----------|---------|------------------------------------------|--------------------------------------------------|-----------------------------------------|
| Glycine   | 56-40-6 | Not applicable                           | Not applicable                                   | Not applicable                          |

## SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects

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| Component | CAS No  | Freshwater Fish                                    | Water Flea        | Freshwater Algae  | Microtox          |
|-----------|---------|----------------------------------------------------|-------------------|-------------------|-------------------|
| Glycine   | 56-40-6 | LC50: > 1000 mg/L,<br>96h static (Oryzias latipes) | No data available | No data available | No data available |

## Persistence and degradability

### **Persistence**

Soluble in water, Persistence is unlikely, based on information available.

## Bioaccumulative potential

Bioaccumulation is unlikely

| Component | log Pow | Bioconcentration factor (BCF) |
|-----------|---------|-------------------------------|
| Glycine   | -3.21   | No data available             |

## Mobility in soil

The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

## Ozone Depletion Potential

| Component | CAS No  | Ozone Depletion Potential |
|-----------|---------|---------------------------|
| Glycine   | 56-40-6 | Not listed                |

## Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

#### **Waste from Residues/Unused Products**

Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

#### **Contaminated Packaging**

Empty containers should be taken to an approved waste handling site for recycling or disposal.

## SECTION 14: TRANSPORT INFORMATION

### Road and Rail Transport

Not Regulated

### IATA

Not regulated

### IMDG/IMO

#### **Marine Pollutant**

Not regulated

No hazards identified

### **Special Precautions for User**

No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

Legend: X - Listed '-' - Not Listed

### **International Inventories**

| Component | CAS No | KECL | TSCA | EINECS | IECSC | DSL | NDSL | PICCS | ENCS | ISHL | AICS |
|-----------|--------|------|------|--------|-------|-----|------|-------|------|------|------|
|-----------|--------|------|------|--------|-------|-----|------|-------|------|------|------|

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|---------|---------|----------|---|-----------|---|---|---|---|---|---|---|
| Glycine | 56-40-6 | KE-01153 | X | 200-272-2 | X | X | - | X | X | X | X |
|---------|---------|----------|---|-----------|---|---|---|---|---|---|---|

| Component | CAS No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Glycine   | 56-40-6 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

| Component | CAS No  | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential |
|-----------|---------|----------|------------------------------|---------------------------|
| Glycine   | 56-40-6 | Listed   | Not applicable               | Not applicable            |

## Korean National Regulations

| Component | CAS No  | Act on Registration and Evaluation of Chemical Substances (K-REACH) | Authorised Chemicals | Existing Substances Subject to Registration |
|-----------|---------|---------------------------------------------------------------------|----------------------|---------------------------------------------|
| Glycine   | 56-40-6 | Annex 1 - KE-01153                                                  | Not applicable       | Not applicable                              |

| Component | CAS No  | Chemical Control Act - Toxic Chemicals | Chemical Control Act - Prohibited Chemicals | Chemical Control Act - Use Restricted Chemicals |
|-----------|---------|----------------------------------------|---------------------------------------------|-------------------------------------------------|
| Glycine   | 56-40-6 | Not applicable                         | Not applicable                              | Not applicable                                  |

| Component | CAS No  | Chemical Control Act - Accident Precaution Chemicals (% in mixtures) | Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures) | Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures) |
|-----------|---------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Glycine   | 56-40-6 | Not applicable                                                       | Not applicable                                                                                 | Not applicable                                                                                         |

| Component | CAS No  | Waste Control Law | Ministry of Environment - CMR risk | Ministry of Environment - Critically Controlled Substance |
|-----------|---------|-------------------|------------------------------------|-----------------------------------------------------------|
| Glycine   | 56-40-6 | Not applicable    | Not applicable                     | Not applicable                                            |

| Component | CAS No  | ISHA - Harmful Agents Subject to Work Environment Monitoring | ISHA - Prohibited substances | ISHA - Substances requiring permission |
|-----------|---------|--------------------------------------------------------------|------------------------------|----------------------------------------|
| Glycine   | 56-40-6 | Not applicable                                               | Not applicable               | Not applicable                         |

| Component | CAS No  | ISHA - Substances subject to control | ISHA - Harmful Agents Requiring Health Examination | ISHA - Permissible Exposure Limits |
|-----------|---------|--------------------------------------|----------------------------------------------------|------------------------------------|
| Glycine   | 56-40-6 | Not applicable                       | Not applicable                                     | Not applicable                     |

| Component | CAS No  | ISHA - Subject to Process Safety Reports (minimum quantity) | ISHA - Threshold Limit Values (TLVs) Chemicals | ISHA - Special management materials |
|-----------|---------|-------------------------------------------------------------|------------------------------------------------|-------------------------------------|
| Glycine   | 56-40-6 | Not applicable                                              | Not applicable                                 | Not applicable                      |

## National Fire Association - Dangerous Substances Minimum quantity requiring a permit

| Component | CAS No  | Class 1 - Oxidising solids | Class 2 - Flammable solid | Class 3 - Spontaneously Combustible Substances and Dangerous Substances When Wet | Class 4 - Flammable liquids | Class 5 - Self-reactive substances | Class 6 - Oxidising liquids |
|-----------|---------|----------------------------|---------------------------|----------------------------------------------------------------------------------|-----------------------------|------------------------------------|-----------------------------|
| Glycine   | 56-40-6 | Not applicable             | Not applicable            | Not applicable                                                                   | Not applicable              | Not applicable                     | Not applicable              |



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## Control Parameters

| Component | CAS No  | Korea      | ACGIH - Biological Exposure Indices |
|-----------|---------|------------|-------------------------------------|
| Glycine   | 56-40-6 | Not listed | Not listed                          |

## US Management Information

OSHA - Occupational Safety and Health Administration

Not applicable

| Component | CAS No  | Specifically Regulated Chemicals | Highly Hazardous Chemicals |
|-----------|---------|----------------------------------|----------------------------|
| Glycine   | 56-40-6 | Not applicable                   | Not applicable             |

CERCLA

Not applicable

| Component | CAS No  | CERCLA Extremely Hazardous Substances RQs | Hazardous Substances RQs | SARA 313 - Threshold Values % |
|-----------|---------|-------------------------------------------|--------------------------|-------------------------------|
| Glycine   | 56-40-6 | Not applicable                            | Not applicable           | Not applicable                |

## GHS Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

Based on available data, the classification criteria are not met.

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

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|                  |                                                    |
|------------------|----------------------------------------------------|
| Prepared By      | Health, Safety and Environmental Department        |
| Creation Date    | 10-Dec-2010                                        |
| Revision Date    | 12-Jun-2024                                        |
| Revision Number  | 2                                                  |
| Revision Summary | New emergency telephone response service provider. |

## MOEL's Public Notice No. 2023-9 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**